

Performance Testing

Date	05 July 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-being (HDI)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the HDI – A Comprehensive Measure of Well-being system performs under different user loads. It measures response time, reliability, and resource usage to ensure smooth operation.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Flask testing
Type of Testing	Load Testing, Performance Testing
Target Module / API	User login, HDI calculation , report generation
Test Environment	Local system (windows , flask server)
Test Date	05-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User login	10	30	Successful login
2	HDI score calculation	20	60	Fast and accurate results
3	Report generation	15	45	Report generated successfully
4	Dashboard loading	25	60	Dashboard loads without delay

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 Sec	Pass	Good Performance
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Within limit
3	Throughput (Req/sec)	50 Req/Sec	55 Req/Sec	Pass	Stable
4	Error Rate	< 1%	0.2%	Pass	Very low errors
5	CPU Utilization	< 80%	58%	Pass	Normal usage
6	Memory Utilization	< 80%	65%	Pass	Stable memory usage

Step 4: Observations & Analysis


Key Findings

- The system responded quickly during testing.
- HDI calculations were accurate and completed successfully.
- Dashboard and reports loaded without noticeable delays.


Bottlenecks Identified

- Minor delay during report generation with higher user load.


Optimization Steps Taken

- Optimized database queries.
- Improved backend processing.
- Reduced unnecessary data loading.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Human Development Index Prediction

Feature 1
 Feature 2
 Feature 3
 Predict

Human Development Index Prediction

Life Expectancy (e.g., 72)
 Expected Schooling (e.g., 12)
 GNI Income (e.g., 15000)
 Predict

Predicted Human Development Index (HDI): 0.6919